Test Cases and Results: End-to-End System Validation Matrix (20/20 PASS)

Phase 6: Comprehensive System Audit Ledger

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1. Comprehensive 20-Point QA Matrix

The system validation matrix verifies all datasets, mathematical calculations, visual assets, and project documentation. All 20 tests passed with 100% integrity.

Table 6.2: System Validation Matrix Results (20 of 20 PASS)

Test	Verification Target	Observed Value	Status
T-01	athletes.csv Existence	Verified on disk	PASS
T-02	athletes.csv Size < 25 MB	21.73 MB (22,788,167 bytes)	PASS
T-03	Modern Era Row Count	157,417 rows (1980-2016)	PASS
T-04	Schema Verification	18 normalized attributes	PASS
T-05	Core Non-Null Check	0 null values in primary attributes	PASS
T-06	Medal Domain Validation	['Gold', 'Silver', 'Bronze', 'No Medal']	PASS
T-07	medal_results.csv Existence	Verified on disk	PASS
T-08	medal_results.csv Rows	3,837 aggregated rows	PASS
T-09	Composite PK Uniqueness	0 duplicate [games_id, noc] tuples	PASS
T-10	Medal Sum Integrity	Total = G + S + B (0 mismatches)	PASS
T-11	Weighted Score Integrity	Score = 3G + 2S + 1B (0 mismatches)	PASS
T-12	Authentic USA Tallies	1,131 Gold, 2,823 Total Medals	PASS
T-13	Host Dataset Existence	host_nation_performance.csv verified	PASS
T-14	Host Surge Rate	86.7% surge rate; +15.4 mean bump	PASS
T-15	Efficiency Dataset Existence	regional_sports_efficiency.csv verified	PASS
T-16	Continental Coverage	7 continental regions mapped	PASS
T-17	Funding Dataset Existence	sports_funding_benchmark.csv verified	PASS
T-18	Funding ROI Correlation	Pearson correlation r = 0.752	PASS
T-19	300 DPI Visualizations	5 charts verified > 2000px width	PASS
T-20	Documentation Suite	7 Phase PDFs + core files present	PASS